

Multiple Choice (20 marks)

1. The lowest possible temperature is absolute zero, at
 - (a) 0 kelvin
 - (b) -273°C
 - (c) Both are the same
 - (d) None of these
2. A magnetic force can act on an electron even when it
 - (a) is at rest
 - (b) moves parallel to magnetic field lines
 - (c) Both of these
 - (d) Neither of these
3. Which of these quantities decreases as the inverse square of distance for distances far from the objects producing the fields?
 - (a) the electric field produced by a finite-length charged rod
 - (b) the electric field produced by an infinitely long charged cylinder
 - (c) the electric field produced by an infinite plane of charge
 - (d) the magnetic field produced by an infinitely long, straight current-carrying wire
4. When a wave moves from shallow water to deep water, the
 - (a) frequency decreases, the wavelength increases, and the speed increases
 - (b) frequency does not change, the wavelength increases, and the speed increases
 - (c) frequency does not change, the wavelength decreases, and the speed decreases
 - (d) frequency does not change, the wavelength increases, and the speed decreases
5. In the absence of air resistance a ball of mass m is tossed upward to reach a height of 20 m. At the 10-m position half way up the net force on the ball is
 - (a) $2mg$
 - (b) mg
 - (c) $mg/2$
 - (d) $mg/4$
6. As more lamps are connected to a series circuit, the current in the power source

- (a) increases
 - (b) decreases
 - (c) remains the same
 - (d) None of these
7. An alarm whose frequency is 400 Hz is dropped out of a third-floor window. The student who drops it measures the frequency with a very sensitive...
- (a) appears higher than 400 Hz and the frequency increase as it falls
 - (b) appears higher than 400 Hz and the frequency decreases as it falls
 - (c) appears lower than 400 Hz and the frequency decreases as it falls
 - (d) appears lower than 400 Hz and the frequency increases as it falls
8. The power required to raise a 100-kg crate a vertical distance of 2 m in a time of 4 s is
- (a) 200 W
 - (b) 500 W
 - (c) 800 W
 - (d) 2000 W
9. During the time an aircraft produces a sonic boom, the aircraft is
- (a) breaking the sound barrier
 - (b) pulling out of a subsonic dive
 - (c) flying faster than sound
 - (d) All of these
10. If you travel at high speed, then compared with your friends who "stay at home," you are
- (a) older.
 - (b) younger.
 - (c) neither younger nor older.
 - (d) longer.

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Compared with a fundamental tone, the frequency of its second harmonic is' is 'half as much'.

2. True or False: The correct answer to 'A model airplane flies slower when flying into the wind and faster with wind at its back. When launched at right angles to the wind a...' is 'greater'.
3. True or False: The correct answer to 'A galvanometer can be calibrated to read electric' is 'voltage'.
4. True or False: The correct answer to 'According to four-dimensional geometry, the angles of a triangle add to 180° ' is 'sometimes'.
5. True or False: The correct answer to 'The "greenhouse gases" that contribute to global warming absorb' is 'more infrared radiation than visible'.
6. True or False: The correct answer to 'An object of mass 2 kg is acted upon by three external forces, each of magnitude 4 N. Which of the following could NOT be the resulting...' is ' 4 m/s^2 '.
7. True or False: The correct answer to 'A hologram best illustrates' is 'superposition'.
8. True or False: The correct answer to 'A big fish swims upon and swallows a small fish that is at rest. Right after lunch, the fattened big fish has a change in' is 'Both'.
9. True or False: The correct answer to 'The Doppler effect occurs when a source of sound moves' is 'Both of these'.
10. True or False: The correct answer to 'According to Hooke's law, if you hang by a tree branch and note how much it bends, then hanging with twice the weight produces' is 'the same bend if the branch doesn't break'.

End of Paper